Early Prediction of Student Academic Performance Using Machine Learning

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Abstract

Researching how students perform in academics is becoming an increasingly important area for researchers in the area of Education Data Mining due to the desire to improve student successes in school and lower the amount of students who drop out of school. Traditionally, academic evaluation systems are very much focused on the final exams of students and fail, in many cases, to identify students who are having a difficult time before they reach the 'end of the road' in terms of academic success. The time delay in recognition of these students makes it more difficult for educators to intervene and help them to be successful. This study introduces a machine learning technique for predicting how students will perform in school based on several academic and behavioral variables, including attendance, internal grades, number of hours studied, and past academic performance. This dataset will undergo preprocessing to ensure the data is clean, normalized, and has had features selected to help improve the accuracy of the resulting model. Multiple machine learning methods, including Logistic Regression, Decision Trees, and Random Forest, will be evaluated using measures of performance for accuracy, precision, recall, and confusion matrix. The results of the experimental evaluation indicate that of the machine learning techniques tested, Random Forest produced the best results with an accuracy of approximately 88%, compared to all other machine learning techniques tested. The results of this study confirm that regular monitoring and evaluation of students will enable educational institutions to identify students who are having difficulty before reaching the end of the academic year while also providing a means by which to support data-driven decision making for improving student success within educational institutions.

Keywords: Machine Learning, Prediction of Student Performance, Educational Data Mining, Random Forest, Classification

I. INTRODUCTION

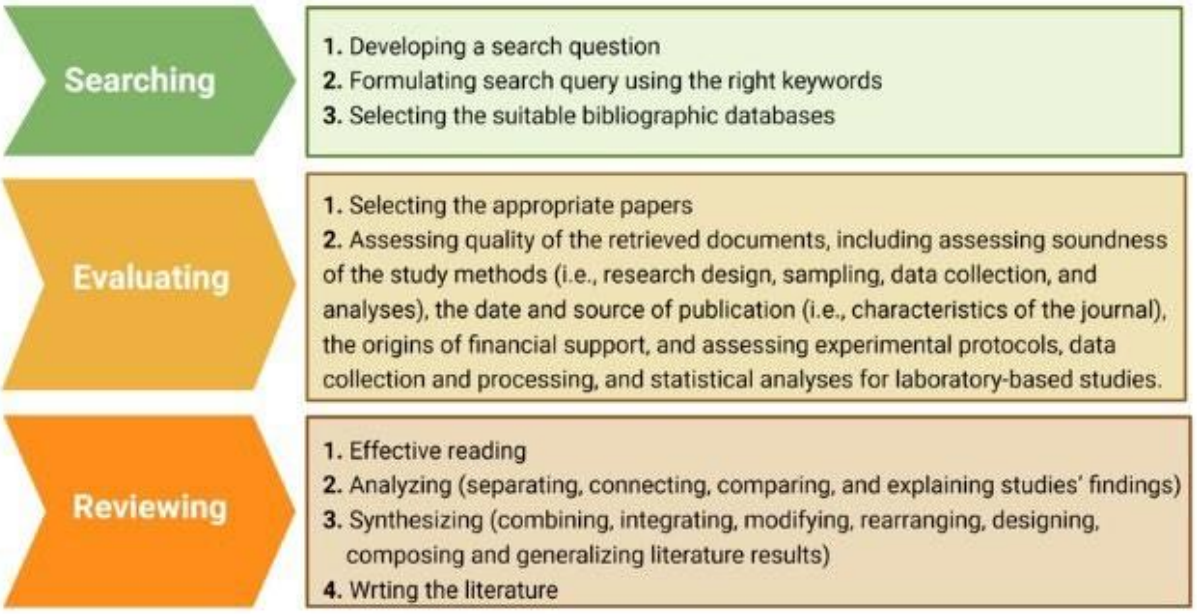
Society has developed through education and academic performance. Academic performance (sometimes referred to as student success) is generally evaluated through an academic performance measurement; however, there are many reasons students struggle, including disengagement from learning, inconsistent class attendance, between study techniques, etc. These problems are impacting student academic performance. Traditional evaluation methods rely primarily on final exams but will not provide educators with timely information about student progress. In contrast, machine learning offers a potential solution by examining historical data for

patterns that can provide an indication of a student's academic performance and developing a predictive academic model. This will enable educators to provide early intervention to improve a student's academic outcome. The objective of this study is to develop a reliable machine learning model for predicting student performance and to identify the major contributing factors.

II. LITERATURE REVIEW

In the past few years, educational data mining has become popular, primarily for predicting student academic performance. A wide range of machine learning techniques has been used by researchers to identify patterns in student data in order to improve educational outcomes. Earlier studies primarily used statistical methods to analyze data, whereas the majority of recent studies have been conducted using machine learning and artificial intelligence methods.

Examples of such studies vary widely; Cortez and Silva [1] were among the first researchers to use data mining on historical academic records and other related data (e.g., grades, study time, and family background) to predict student success in future academic endeavors. Their results demonstrated that previous academic performance is a reliable predictor of success in future academic endeavors. Romero and Ventura [2] provided an extensive review of the educational data mining literature and noted that preprocessing, feature selection, and model evaluation are critical to improving the accuracy of predictions made by educational data mining. Kotsiantis et al. [3] compared the effectiveness of different classification algorithms, including Decision Trees, Naïve Bayes, and Neural Networks in categorizing student academic performance. They found that while easier to understand, simpler models do not frequently produce accurate predictions. On the other hand, more sophisticated models, such as Random Forest [4], tend to provide the most accurate predictions. More recent studies have focused on developing hybrid and deep learning models to make more accurate predictions; however, they require large datasets and a substantial amount of computational power.



III. SYSTEM ARCHITECTURE

This system will be designed in several distinct phases. The first phase includes data collection, which will consist of compiling individual student data (e.g., attendance, internal mark assessment for that subject, study hours and past performance) from academic records. The next phase is pre-processing by addressing any missing data, inconsistencies, and the application of normalization to ensure data integrity. Feature selection will then be performed on the data set to isolate and identify the most relevant features for the purpose of predicting student performance. Next, machine learning algorithms will be trained using the procured pre-processed data, and predictions will be made based on those trained models. Finally, the models will be evaluated using various performance metrics, and the best performing model will be selected for final deployment within the system.

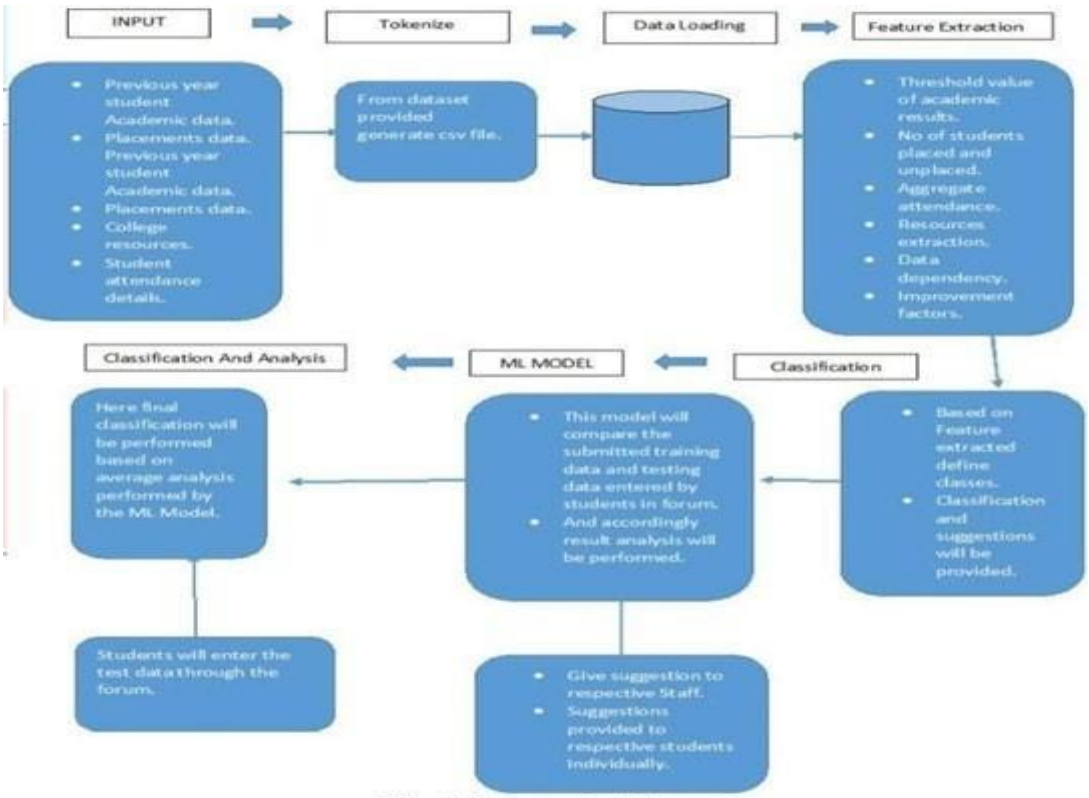
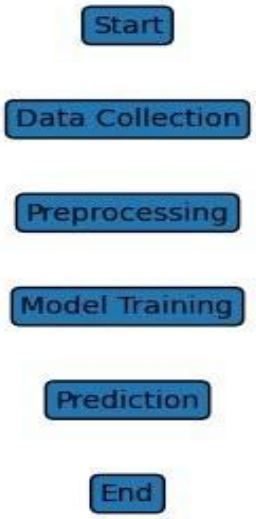


Fig. 1 System Architecture

IV.METHODOLOGY

For this study's methodology we will follow four main steps: data collection, preprocessing, model training, and evaluation. A description of the methods and resources used will be documented for each step. A total of 2 experiments will be conducted (1 experiment per machine learning algorithm). The methods used are Logistic Regression (LR), Decision Tree (DT), and Random Forest (RF). Logistic Regression is used to classify whether a grouping is a binary classification; Decision Tree provides guidance on a structured way to make decisions; Random Forest is an ensemble of multiple Decision Trees and combines them in order to obtain the most accurate.results.



V. EXPERIMENTAL SETUP

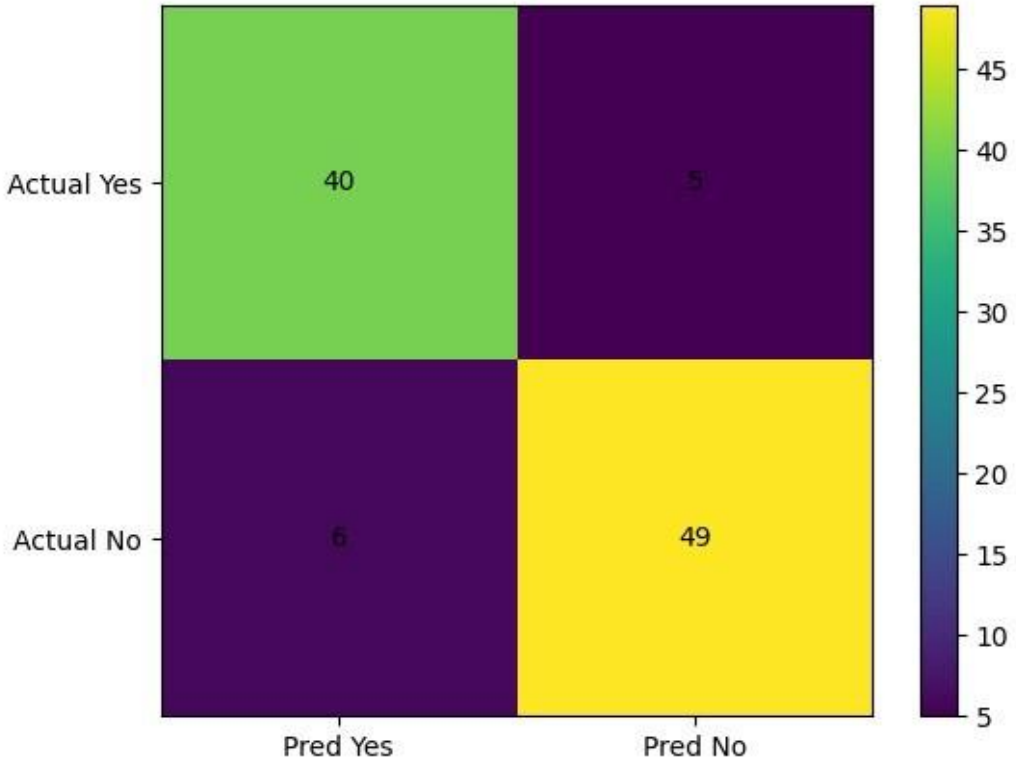
For our experiments we will use Python and machine learning libraries such as Scikit-learn for modeling and data preparation. The data set used will be a total dataset of the following: Training (Total of 80% of data) and Test (Total of 20% of data). The Training data will be used for model training, while the Test data will be used to evaluate the overall performance of the machine learning algorithm. Evaluation/calculation of overall performance of machine learning algorithms will include evaluation metrics such as accuracy, precision, recall, and confusion matrix.

Table 4.1: Descriptive Statics

Variable	Minimum	Maximum	Mean	Std. Deviation
Attendance	40	100	76	12
Internal Marks	10	50	32	8
External Marks	20	100	65	15
Study Time(hrs.)	0	5	2.5	1.1

VI. RESULTS AND DISCUSSION

Based on the results obtained from the testing of the three machine-learning algorithms, Random Forest gained the highest overall accuracy and performance from each individual classification. Decision Trees ranked second based on their ability to classify; and printed out through metrics, whereas Logistic Regression was the lowest-performing algorithm of the three algorithms. Based on the Confusion Matrix, Random Forest appears to have the most accurate classification with the least amount of errors between classification; and based on the analysis of the Features Importance, Attendance and Internal Marks were the 2 most relevant criteria for a student's academic performance.



VII. APPLICATIONS

Machine learning has many uses for predicting student success in education today. One common reason by having this program, allows teachers to notice when a student is struggling early on, so corrections can be implemented, like providing additional assistance or having study groups; another application created with this machine learning system creates personalized environments. Traditionally, teachers use one method with all their students; this program gives teachers the ability to modify their techniques to fit each student's unique style of learning. For example, students who receive an indication of having difficulty in subjects would now be given an additional amount of resources and have them focus instruction on those subjects more directly, leading to a better outcome for them. The system's contribution to the educational

system includes figuring out the data to help administratively develop the school and make decisions based upon looking at trends that exist in the student population. The data these trends provide will ultimately help in designing curriculum, utilizing resources and providing better teacher techniques. The machine learning program can also be used with an e-learning system to give students immediate feedback.

This allows for tracking individual progress and adjusting study habits to help improve learning. In addition to academic application, career guidance and counselling are enhanced through this system. By evaluating student performance, educators can identify areas of strength or weakness in order to provide better support regarding future career choices. The complete support offered to students throughout their academic careers through the use of this system creates an all-inclusive method of providing assistance to students throughout their academic career.

VIII. ADVANTAGES

Compared to traditional methods of evaluating student performance, the machine learning-based student performance evaluation system being proposed offers numerous advantages. The first major benefit of such a system is its ability to predict student outcomes early enough for educators to take corrective action before the student fails. In contrast, traditional evaluation methods typically rely solely on final exams for evaluating student performance; whereas a machine learning-based evaluation system enables educators to track their students' performance continuously and provide real-time feedback.

Another major benefit of using machine learning for student performance evaluation is the level of accuracy and reliability that comes from using advanced machine learning algorithms. Specifically, there is a great deal of accuracy and reliability gained from employing ensemble methods such as random forests that combine multiple models to arrive at predictions. By combining multiple models, the machine learning-based evaluation system reduces the likelihood of overfitting and improves generalization.

Larger educational institutions will benefit from additional advantages associated with scalability and organization as they will be able to accommodate higher volumes of data due to the increased number of students who are currently enrolled in their institutions as well as the exponential growth of digital learning platforms that produce massive amounts of data on a daily basis. Since the proposed machine learning-based student performance evaluation system will have the capability to analyze this vast amount of data quickly, it will provide institutions with valuable insights about their student body.

Finally, transparency and objectivity will increase as a result of a machine learning-based evaluation system compared with traditional assessment methods due to the objective nature of the data analysis performed by a machine learning system, while traditional assessment techniques may depend upon the evaluator's subjective impressions about a student. The machine learning-based evaluation system is relatively easy to implement and can be integrated into existing educational systems, allowing for streamlined implementation processes.

It is a feasible solution for improving academic results due to its ability to be created through a variety of generally available tools and technologies and integrated into pre-existing educational institutions.

IX. LIMITATIONS AND FUTURE WORK

Despite its numerous advantages, the proposed system has certain limitations that must be While there are many good points about this method it does have some limitations as well. One major limitation is how well the data is gathered for training the Machine Learning Model. The quality of your data will determine how accurate the predictions you make will be. Finding appropriate data in the form you need can be difficult due to constraints on gathering quality data for real-world applications. Another issue that arises from this method is that there are other factors that will affect how well a student does besides academic performance. Examples include psychological, social and/or economic issues that could potentially affect a student's overall success but cannot be processed into the models used. Using complicated algorithms also creates an obstacle when trying to understand predictions made for academic success; even though Random Forest Models give higher accuracy in predictions than Decision Tree Models, they will be harder for educators to understand and therefore make the predictions difficult to put into practice. Future improvements can be achieved by using Deep Learning to provide even more accurate results as a result of better training data.Another area of future research is the inclusion of additional features such as behavioral data, psychological assessments, and socio-economic factors. This will provide a more comprehensive understanding of student performance and improve the overall effectiveness of the system. Developing user-friendly interfaces and web-based applications can also enhance the usability of the system. The addition of real time data can also allow for more accurate predictions and the ability to continuously monitor students' performance.

Future research could also involve using other variables to predict students' chance of success, like using behavioural data, psychological assessments and socio-economic factors, to provide a more detailed understanding of both the student's performance and increasing the overall trigger effectivety. Finally, developing user-friendly interfaces, & web based applications would increaseig the effectiveness of the system.

X. CONCLUSION

Overall, this paper provides an innovative and complete solution for predicting how students will do academically through the use of machine learning methods. This study was able to confirm that machine learning methods can accurately assess student data and establish patterns that can be used to predict future performance. Based on the results from this study, Random Forest algorithm produced the highest prediction accuracy and the best overall model among those evaluated in this research.

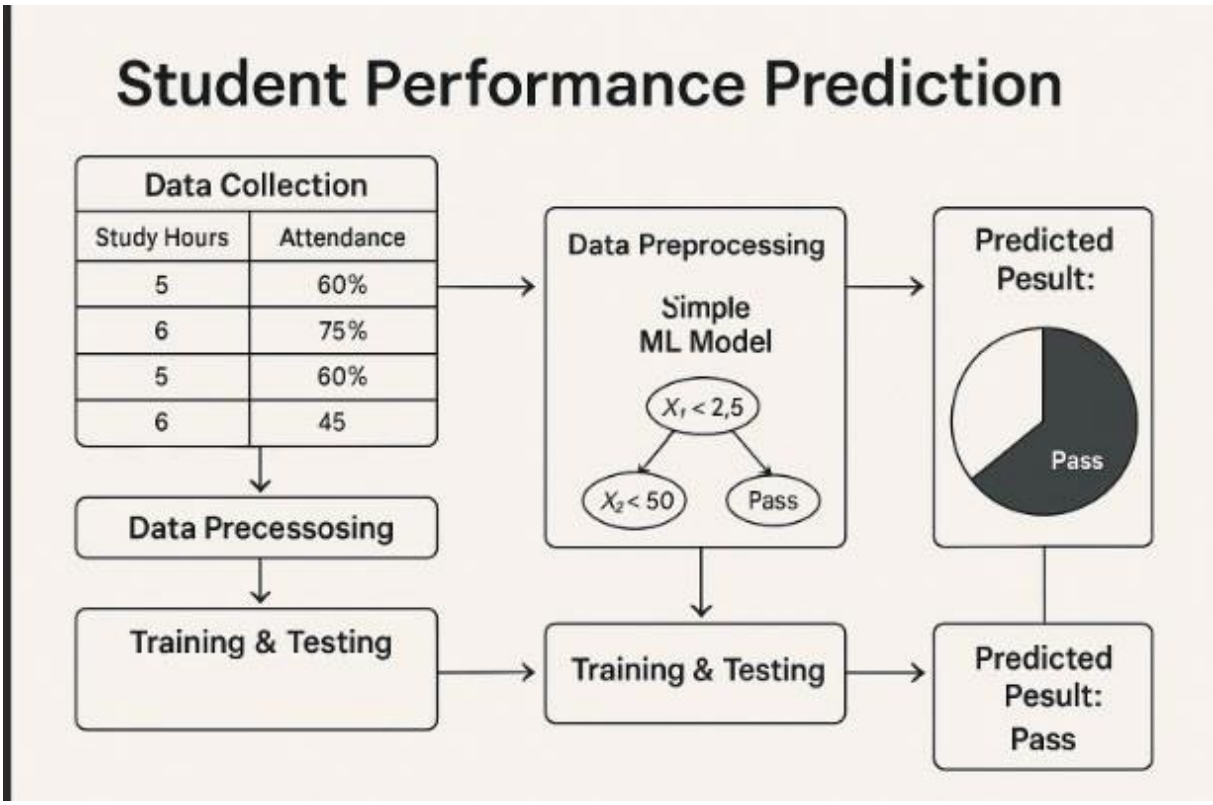
The findings from our proposed system have major implications for the education field, as it will provide a means for early identification of at-risk students and allow educators to make targeted

interventions to help improve academic success through timely identification. A proactive approach will not only be beneficial to individual students, but also improve the maintenance of an overall high quality of education.

The results of this study also provide educators with valuable information about the importance of specific variables such as attendance, study habits, and internal assessment scores, all which aid in determining a student's level of academic success. By understanding the results of this study, educators should be able to utilize the findings to develop effective teaching methods and strategies that will help improve the level of student engagement they are encountering with their students.

While there are limitations with the current proposed system, this research provides the basis for future research in the area of education data mining. With some additional enhancements and modifications to incorporate additional features, the proposed system can be enhanced and made to be more useful in the real world.

The overall conclusion of this research is that the use of machine learning could revolutionize the way we educate our children by increasing their likelihood of being successful in school. With access to large sets of data, educational institutions will be able to build much more efficient and individualized educational environments than what exists today. Ultimately, this should result in improved academic performance and lower rates of dropout from school.



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